

BARRAGE TEMISCAMINGUE, QC, Canada

WMO: 717320

Lat: **46.7097N**Lon: **79.1010W**Elev: **595**StdP: **14.38**Time Zone: **-5.00 (NAE)**Period: **97-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.9	-9.7	-21.1	1.8	-14.2	-15.8	2.4	-8.5	18.5	10.4	17.1	9.8	5.1	10	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	85.4	68.7	82.3	67.1	79.2	65.4	71.8	80.3	70.1	77.6	68.3	75.1	7.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.2	109.8	75.1	67.5	103.5	73.4	65.9	97.7	71.9	35.9	80.5	34.4	77.5	33.0	75.4	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.1	14.1	12.4	-21.4	90.2	5.2	2.8	-25.1	92.2	-28.2	93.9	-31.1	95.5	-34.9	97.5		
(5)			-21.7	75.5	5.1	2.6	-25.4	77.4	-28.4	79.0	-31.3	80.5	-35.0	82.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.6	12.7	15.8	25.5	39.0	52.2	61.5	67.3	65.8	58.4	45.8	33.4	20.5
	DBStd	21.01	12.98	11.33	11.42	8.22	7.62	6.62	5.30	5.32	7.35	7.98	8.94	11.35
	HDD50	4891	1156	958	759	340	65	3	0	0	12	184	500	914
	HDD65	8783	1621	1378	1223	780	402	143	35	55	223	596	948	1379
	CDD50	1836	0	0	1	10	133	349	535	489	263	54	2	0
	CDD65	252	0	0	0	0	5	38	105	79	24	1	0	0
	CDH74	2206	0	0	2	6	158	424	884	555	168	9	0	0
	CDH80	533	0	0	0	0	35	100	238	126	34	0	0	0
Wind	WSAvg	5.7	6.2	6.2	6.5	6.5	5.8	4.9	4.7	4.6	4.9	5.7	6.2	6.1
Precipitation	PrecAvg	36.3	2.5	2.0	2.4	2.8	3.1	3.2	3.6	3.3	3.6	3.7	3.3	2.7
	PrecMax	43.4	4.3	4.2	8.5	5.1	5.5	6.0	7.4	5.8	6.2	6.8	5.9	5.6
	PrecMin	27.1	0.9	0.8	0.0	1.0	0.7	1.1	1.1	0.9	1.7	1.9	1.4	1.2
	PrecStd	4.3	0.9	0.9	1.6	1.0	1.3	1.3	1.7	1.3	1.3	1.5	1.1	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.5	44.6	59.0	73.4	84.4	86.8	90.1	88.1	84.7	74.3	59.0	47.1
		MCWB	41.7	39.4	46.3	56.9	65.3	68.9	71.4	69.9	70.2	61.2	51.8	44.0
	2%	DB	37.7	38.6	50.3	64.5	78.2	83.0	85.8	83.5	78.6	67.9	53.0	41.7
		MCWB	36.0	35.2	41.3	49.3	60.1	66.8	69.0	68.3	67.2	59.3	48.7	39.7
	5%	DB	34.8	35.4	44.9	58.5	73.5	79.2	82.3	80.0	74.6	62.4	49.1	38.1
		MCWB	33.3	32.9	38.6	46.2	57.4	64.8	67.3	66.4	64.5	55.8	45.9	36.3
	10%	DB	32.0	32.4	40.4	52.9	68.1	75.0	79.0	76.6	70.8	58.1	45.5	35.5
		MCWB	30.4	30.5	35.6	42.8	55.3	62.4	65.4	65.1	63.4	53.7	42.8	33.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.4	40.7	50.1	58.5	68.5	72.4	75.4	73.2	72.2	64.5	54.0	44.8
		MCDB	43.3	43.5	53.2	68.5	80.5	82.5	85.8	82.3	80.4	69.7	57.6	45.8
	2%	WB	36.4	35.8	43.2	52.0	63.7	70.1	72.1	70.8	69.4	60.1	49.4	39.9
		MCDB	37.4	38.1	48.4	61.5	72.7	79.0	80.1	78.9	74.8	66.1	52.2	41.3
	5%	WB	33.4	33.1	39.1	47.1	60.0	67.3	70.1	69.2	67.0	57.2	46.1	36.5
		MCDB	34.6	35.0	43.8	56.1	69.1	74.9	77.3	76.2	72.3	61.7	48.9	38.0
	10%	WB	30.4	30.4	36.1	43.8	56.5	64.9	68.2	67.4	64.4	53.7	42.9	34.1
		MCDB	31.8	32.5	39.5	51.6	65.1	71.7	75.2	73.3	69.3	58.1	45.5	35.4
Mean Daily Temperature Range	5% DB	MDBR	16.4	18.0	19.3	19.6	23.5	20.9	19.9	18.1	17.7	13.8	11.5	12.8
		MCDBR	14.9	17.3	23.5	31.3	33.4	27.1	24.3	22.9	22.6	21.3	17.2	12.5
		MCWBR	14.1	14.2	15.5	17.5	16.4	12.6	10.2	10.0	11.8	13.0	13.4	11.6
	5% WB	MCDBR	14.5	15.7	21.2	27.9	28.6	22.6	20.9	18.8	19.4	19.4	16.0	12.2
		MCWBR	14.1	13.6	14.8	17.4	16.2	12.3	10.4	9.7	11.6	13.1	13.4	11.6
Clear-Sky Solar Irradiance	taub		0.270	0.286	0.322	0.358	0.369	0.400	0.402	0.394	0.372	0.338	0.304	0.272
	taud		2.317	2.252	2.268	2.340	2.391	2.333	2.351	2.369	2.409	2.492	2.501	2.383
	Ebn at Noon		262	280	285	285	284	275	272	270	266	260	250	249
	Edh at Noon		23	31	36	37	36	39	38	36	31	25	20	20
All-Sky Solar Radiation	RadAvg		404	705	1168	1445	1652	1789	1776	1515	1151	688	373	298
	RadStd		57	93	121	118	144	130	136	98	123	84	44	27